What you will learn in this chapter

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- How to compute complexity through **compression**
- How to compare **algorithmic information** with Shannon's information
- How to **detect languages** through joint compression
- How to use the Web to compute **meaning similarity**

Detailed outline Chapter (2)

Measuring Information through compression

Compressing objects

We defined the complexity of an object as the length of its shortest description. This most compact description can be seen as the result of a *compression*. In this chapter, we present some basic properties of compression and some basic compression algorithms.

Using compression as a tool

An object is compressible if it contains some regularities. We will show that this property can be used to perform some simple tasks, such as language recognition: two texts written in the same language contain similar patterns and can be easily co-compressed.

Complexity and frequency

We explore the link between simplicity and frequency. We expect simple objects, such as words or concepts, to be more frequent. We verify this expectation by observing word frequencies in language and on the Web.